KENYA INSTITUTE OF CURRICULUM DEVELOPMENT

A skilled and Ethical Society

PRIMARY SCHOOL EDUCATION CURRICULUM DESIGN

MATHEMATICAL ACTIVITIES

GRADE 2

First Published in 2017

Revised 2024

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ISBN: 978-9914-43-079-0

Published and printed by Kenya Institute of Curriculum Development

FOREWORD

The Government of Kenya (GoK) is committed to ensuring that policy objectives for Education, Training and Research meet the aspirations of the Constitution of Kenya 2010, the Kenya Vision 2030, National Curriculum Policy 2018, the United Nations Sustainable Development Goals (SDGs) and the regional and global conventions to which Kenya is a signatory. Towards achieving the mission of Basic Education, the Ministry of Education (MoE) has successfully and progressively rolled out the implementation of the Competency Based Curriculum (CBC) at Pre-Primary, Primary and Junior School levels.

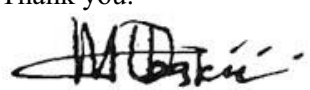
The Kenya Institute of Curriculum Development (KICD) reviewed the curriculum and rationalised the number of learning areas in 2024. The review and rationalisation process was informed by several factors, among them, the recommendations of the Presidential Working Party on Education Reforms (PWPER) and reports of the continuous curriculum monitoring and evaluation activities.

The reviewed curriculum designs build on competencies attained earlier by learners. The designs prepare the learner for smooth transition to the next level. The designs will also afford the learner opportunities for developing requisite competencies and enable them to interact with other people and the environment around them.

The key components of the curriculum designs include the National Goals of Education, the essence statement, general and specific learning outcomes as well as the strands and sub strands. Suggested learning experiences, key inquiry questions, core competencies, Pertinent and Contemporary Issues (PCIs), values and assessment rubrics are also outlined in the curriculum designs.

It is expected that all Government agencies and other stakeholders in Education will use the designs to plan for the effective and efficient implementation of the Competency Based Curriculum.

Thank you.



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PREFACE

The Ministry of Education (MoE) rolled out the Competency Based Curriculum (CBC), nationally in 2019, following a national convention in 2017 where the *Basic Education Curriculum Framework* (BECF) was adopted by stakeholders and a national pilot of the curriculum in the Early Years of Education (EYE) in 2018. According to the UNESCO IBE requirements, a curriculum should be reviewed every five years. So, the review of CBC was due from 2023. In view of this, the reviewed curriculum designs will enhance the implementation of CBC since it corporates the lessons learnt from the implementation of CBC so far.

Consistent periodical review of the curriculum is also critical in the realisation of the Vision and Mission of the on-going curriculum reforms as enshrined in the Sessional Paper No. I of 2019 whose title is: *Towards Realizing Quality, Relevant and Inclusive Education and Training for Sustainable Development* in Kenya. The Sessional Paper explains the shift from a content-focused curriculum to a focus on producing an engaged, empowered and ethical citizen.

Therefore, the reviewed curriculum designs will facilitate the inculcation of core competencies in CBC, which are identified as: communication and collaboration, critical thinking and problem solving, creativity and imagination, citizenship, digital literacy, learning to learn and self-efficacy.

The curriculum designs provide suggestions for interactive and differentiated learning experiences linked to the various strands and sub strands and other aspects of the CBC. The designs also outline suggested learning resources and varied assessment techniques. It is expected that the use of these designs will lead to enhanced learning outcomes at various levels, prepare the learner for smooth transition to subsequent grades and make learning enjoyable.

The MoE requests all stakeholders to keep giving feedback on the curriculum designs to inform the review during the next cycle.

Thank you.

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ACKNOWLEDGEMENT

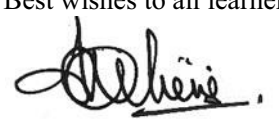
The Kenya Institute of Curriculum Development (KICD) Act Number 4 of 2013 (Revised 2019) mandates the Institute to develop and review curricula and curriculum support materials for basic and tertiary education and training. The curriculum development process is guided by research, international best practices as well as stakeholder engagement. The Institute conceptualised the Competency Based Curriculum (CBC) in consultation with the Ministry of Education and other stakeholders. According to the *Basic Education Curriculum Framework* (KICD, 2017) the conceptualisation of CBC was informed by 21st Century learning needs, the Constitution of Kenya 2010, the Kenya Vision 2030, the East African Community Protocol, the International Bureau of Education (IBE) Guidelines and the United Nations Sustainable Development Goals (SDGs).

KICD is funded by the Kenya Government to discharge its mandate. The institute also receives support from development partners targeting specific programmes. The reviewed curriculum designs were developed with the support of the World Bank through the Kenya Primary Education Equity in Learning Programme (KPEELP) - a project coordinated by MoE. KICD wishes to most sincerely thank the Government of Kenya, through the MoE and other development partners. More specifically, KICD appreciates the Cabinet Secretary - MoE and the Principal Secretary – State Department of Basic Education,

Additionally, the Institute expresses gratitude to all the KICD staff members, teachers, university lecturers, MoE staff, Sem i-Autonomous Government Agencies (SAGAs) and representatives of various stakeholders; among others, for their contributions to the development of the reviewed curriculum designs. Finally, KICD acknowledges the Chief Executive Officers of the Teachers Service Comm ission (TSC) and the Kenya National Examinations Council (KNEC) as well as the KICD Council for supporting the curriculum review process.

May God bless all the individuals and respective institutions who in one way or another supported the curriculum review process. Indeed, these designs will effectively guide the implementation of the CBC at Primary level, thereby preparing the learner to transition to the Junior School.

Best wishes to all learners and curriculum implementers.



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NATIONAL GOALS OF EDUCATION

1. **Foster nationalism, patriotism, and promote national unity**

   Kenya's people belong to different communities, races and religions and should be able to live and interact as one people. Education should enable the learner acquire a sense of nationhood and patriotism. It should also promote peace and mutual respect for harmonious co-existence.

2. **Promote social, economic, technological and industrial needs for national development**

   Education should prepare the learner to play an effective and productive role in the nation.

   a) **Social Needs**

      Education should instil social and adaptive skills in the learner for effective participation in community and national development.

   b) **Economic Needs**

      Education should prepare a learner with requisite competences that support a modern and independent growing economy. This should translate into high standards of living for every individual.

   c) **Technological and Industrial Needs**

      Education should provide the learner with necessary competences for technological and industrial development in tandem with changing global trends.

3. **Promote individual development and self-fulfilment**

   Education should provide opportunities for the learner to develop to the fullest potential. This includes development of one's interests, talents and character for positive contribution to the society.

4 Promote sound moral and religious values

Education should promote acquisition of national values as enshrined in the Constitution. It should be geared towards developing a self-disciplined and ethical citizen with sound moral and religious values.

5. Promote social equity and responsibility

Education should promote social equity and responsibility. It should provide inclusive and equitable access to quality and differentiated education; including learners with special educational needs and disabilities. Education should also provide the learner with opportunities for shared responsibility and accountability through service learning.

6. Promote respect for and development of Kenya's rich and varied cultures

Education should instil in the learner appreciation of Kenya's rich and diverse cultural heritage. The learner should value own and respect other people's culture as well as embrace positive cultural practices in a dynamic society.

7. Promote international consciousness and foster positive attitudes towards other nations

Kenya is part of the interdependent network of diverse peoples and nations. Education should therefore enable the learner to respect, appreciate and participate in the opportunities within the international community. Education should also facilitate the learner to operate within the international community with full knowledge of the obligations, responsibilities, rights and benefits that this membership entails.

8. Good health and environmental protection

Education should inculcate in the learner the value of physical and psychological well-being for self and others. It should promote environmental preservation and conservation, including animal welfare for sustainable development.

LESSON ALLOCATION AT LOWER PRIMARY

S/No	Learning Area	Number of Lessons Per Week
1.	Indigenous Language Activities	2
2.	Kiswahili Language Activities / Kenya Sign Language Activities	4
3.	English Language Activities	5
4.	Mathematical Activities	5
5.	Religious Education Activities	3
6.	Environmental Activities	4
7.	Creative Activities	7
8.	Pastoral/Religious Instruction Programme	1*
Total		**31**

LEVEL LEARNING OUTCOMES FOR PRIMARY SCHOOL EDUCATION

By the end of Primary Education, the learner should be able to:

a) Use verbal and or non-verbal cues to convey information in varied contexts.
b) Demonstrate mastery of number concepts to solve problems in day to day life.
c) Use appropriate social skills, moral and religious values to positively impact the society.
d) Develop individual talents and interests for self-efficacy.
e) Make informed decisions as local and global citizens of a diverse, democratic society in an interdependent world.
f) Devise innovative strategies for environmental conservation and sustainability.
g) Apply digital literacy skills for learning and enjoyment.
h) Appreciate the country's rich and diverse cultural heritage for harmonious living.

MATHEMATICAL ACTIVITIES
GRADE 2

ESSENCE STATEMENT

Mathematics is a learning area that involves computation in numbers and arithmetic, working with shapes, understanding spatial relationships, and processing information in the form of data. It plays a crucial role in driving a country's economic development. By learning mathematics, learners develop an understanding of numbers, logical thinking skills and problem-solving abilities. These skills are essential not only in business but also in the social and political spheres. At this level, mathematics builds on the competencies acquired by the learner in the early years of education. It also enhance strengthens their numeracy skills, which serve as a foundation for STEM at higher levels of education. Importantly, mathematics can also a subject of enjoyment and excitement, offering learners opportunities for creative work and fun.

SUBJECT GENERAL LEARNING OUTCOMES

By the end of Primary Education, the learner should be able to:

1. demonstrate mastery of number concepts by working out problems in day-to-day life.
2. apply measurement skills to find solutions to problems in a variety of contexts.
3. apply properties of geometrical shapes and spatial relationships in real-life experiences.
4. apply data handling skills to solve problems in day-to-day life.
5. analyse information using algebraic expressions in real-life situations.
6. apply mathematical ideas and concepts to other learning areas or subjects and in real-life contexts.
7. develop confidence and interest in mathematics for further learning and enjoyment.
8. develop values and competencies for a cohesive harmonious living in the society.
9. manage pertinent and contemporary issues for enhanced interpersonal relationships

SUMMARY OF STRANDS AND SUB-STRANDS

Strands	Sub-Strands	Suggested Number of Lessons
1.0 Numbers	1.1 Pre-Number Activities	20
	1.2 Whole Numbers	25
	1.3 Addition	25
	1.4 Subtraction	20
2.0 Measurements	2.1 Length	10
	2.2 Mass	10
	2.3 Capacity	12
	2.4 Time	8
	2.5 Money	8
3.0 Geometry	3.1 Lines	6
	3.2 Shapes	6
Total Number of Lessons		**150**

NOTE:
The suggested number of lessons per sub-strand may be less or more depending on the context.

STRAND 1.0: NUMBERS

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Numbers	**1.1 Number Concept** (8 lessons) • *Numbers in symbols* • *Number games*	By the end of the sub-strand, the learner should be able to: a) identify numbers 1 to 100 in symbols in different situations, b) represent numbers 1 to 100 using concrete objects from the environment, c) play number games using number cards or digital devices, d) appreciate the use of numbers in real-life situations.	The learner is guided to: ● recognise and read numbers 1 to 100 from number cards or charts, ● collect safe concrete objects from the environment, ● count concrete objects of given numbers in symbols, ● match a group of objects to their number value, ● in turns, discuss, choose, and play number games in turns using number cards or digital devices.	How can we represent numbers using objects?
Core Competencies to be developed: ● Digital Literacy: learner uses digital devices to play number games. ● Learning to Learn: learner discovers ways of representing numbers as they match a group of objects to their number value.				
Values: ● Unity: learner respects peers' opinions as they in turn, discuss, choose, and play number games. ● Responsibility: learner observes safety practises as they collect concrete objects for learning from the environment.				

Pertinent and Contemporary Issues (PCIs): ● Social Cohesion: learner discusses, chooses, and plays number games in turns. ● Safety issues: learner observes safety as they collect concrete objects for learning from the environment.
Link to other learning areas: The learner relates the concept of using concrete objects from the environment to represent numbers to the concept of resources in the environment in Environmental Activities.

Strand	Sub-strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Numbers	**1.2 Whole Numbers** (20 lessons) • *Counting numbers forward and backward* • *Place value* • *Missing numbers*	By the end of the sub-strand, the learner should be able to: a) count numbers forward up to 100 in different situations, b) count numbers backward from the number 50, c) read and write numbers 1 to 100 in symbols in different situations, d) identify the place value of numbers in ones and tens, e) read and write numbers 1 to 20 in words, f) work out missing numbers in number patterns up to 100,	The learner is guided to: • count numbers forward up to 100 starting from any point, • count numbers backward from 50 starting from any point, • recognise and read numbers 1 to 100 in symbols from number cards or charts, • name and write the numbers in the place value of ones and tens, • discuss the place value of digits written on the number cards, • read and write numbers 1 to 20 in words, • work out missing numbers in number patterns up to 100, • make number patterns and share with peers, • play games involving whole numbers using digital devices or other resources, • improvise place value apparatus by use of place value tins and pockets from locally available materials,	How do we get the next number in a number pattern?

		g) appreciate number patterns in playing number games.	• play a game of putting number cards in place value tins or pockets (ones and tens) according to the place value of digits.	

Core Competencies to be developed:

- Creativity and Imagination: learner improvises place value apparatus by use of place value tins and pockets from locally available materials.
- Communication and Collaboration: the learner discusses the place value of digits written on the number cards.

Values:

- Unity: the learner, in turn, plays a game of putting number cards in place-value tins or pockets according to the place value of digits.
- Responsibility: learner observes safety precautions as they use locally available materials to improvise place value tins and pockets.

Pertinent and Contemporary Issues (PCIs):

- Effective Communication: learner reads and writes numbers 1 to 100 in symbols in different situations.
- Creative thinking: learner improvises place value tins and pockets from locally available materials.

Link to other learning areas:

- The learner relates the skills used in reading and writing numbers in symbols to reading and writing skills in English Language Activities.
- The learner relates the skills used in making number patterns to the concept of patterns in Creative Activities.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Numbers	**1.3 Addition** (20 lessons) • *Addition of numbers* • *Missing numbers*	By the end of the sub-strand, the learner should be able to: a) add a 2-digit number to a 1-digit number without and with regrouping with the sum not exceeding 100, b) add a 2-digit number to a 2-digit number without and with regrouping, with the sum not exceeding 100, c) add two multiples of 10 whose sum does not exceed 100, d) work out missing numbers in patterns involving the addition of whole numbers up to 100, e) appreciate the addition of numbers in real-life situations.	The learner is guided to: • write additional sentences given in horizontal form into vertical form according to place value, • add a 2-digit number to a 1-digit number without regrouping, • use number cards to add a 2-digit number to a 1-digit number with regrouping, • in teams, discuss and come up with different ways of adding two 2-digit numbers without and with regrouping, • add 2 multiples of ten whose sum does not exceed 100, • work with peers and make patterns using numbers up to 100 and share with others.	1. How are horizontal addition sentences written vertically? 2. When do we regroup during addition?

			• play games with peers involving addition using digital devices or other resources,	

Core Competencies to be developed:

- Communication and Collaboration: The learner speaks and listens to peers as they discuss different ways of adding two 2-digit numbers without and with regrouping.
- Learning to Learn: learner discovers ways of presenting addition as they write addition statements in horizontal and vertical forms.

Values:

- Social Justice: learner accommodates others as they play games involving addition.
- Unity: The learner discusses and comes up with different ways of adding two 2-digit numbers without and with regrouping.

Pertinent and Contemporary Issues (PCIs):

- Critical thinking: learner works out missing numbers in patterns involving addition.
- Friendship formation: learner plays games with peers involving addition using different resources.

Link to other learning areas:

- The learner relates the skills used in making patterns to the concept of patterns in Creative Activities.
- The learner relates the skills used in writing additional sentences in horizontal and vertical forms to functional writing in English Language Activities.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Numbers	**1.4 Subtraction** (20 lessons) • *Subtraction of numbers* • *Missing* numbers	By the end of the sub-strand, the learner should be able to: a) subtract a 1-digit number from a 2-digit number without regrouping, b) subtract a 2-digit number from a 2-digit number without and with regrouping, c) subtract a lower multiple of 10 from a higher multiple of 10, d) work out missing numbers in patterns involving subtraction up to 100, e) appreciate the subtraction of numbers in real-life situations.	The learner is guided to: ● subtract a 1-digit number from a 2-digit number without regrouping, ● subtract a 2-digit number from a 2-digit number without regrouping using place value apparatus, ● subtract a 2-digit number from a 2-digit number with regrouping using place value apparatus, ● subtract lower multiples of 10 from higher multiples of 10 using number cards, ● in teams, discuss and work out missing numbers in patterns involving subtraction up to 100.	How do you work out missing numbers in number patterns involving subtraction?

Core Competencies to be developed:

- Learning to Learn: the learner discovers steps of subtracting a 2-digit number from a 2-digit number with regrouping using place value apparatus.
- Critical thinking: learner discusses and works out missing numbers in patterns involving subtraction up to 100.

Values:

- Unity: learner collaborates as they discuss and work out missing numbers in patterns involving subtraction up to 100.
- Social Justice: The learner accommodates peers as they discuss and work out missing numbers in patterns involving subtraction.

Pertinent and Contemporary Issues (PCIs):

- Social Cohesion: learner jointly with others discusses and works out missing numbers in patterns involving subtraction up to 100.
- Critical thinking: learner subtracts a 2-digit number from a 2-digit number with regrouping using place value apparatus.

Link to other learning areas:

The learner relates the skills used in discussion to speaking and listening skills in English Language Activities.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Numbers	**1.5 Multiplication** (12 lessons) • *Multiplication of numbers* • *Multiplication sign*	By the end of the sub-strand, the learner should be able to: a) represent multiplication as repeated addition using numbers 1, 2, 3, by 2 and 3, b) write repeated addition sentences as multiplication using '×' sign, c) multiply 1-digit numbers by 1, 2, 3, 4 and 5, d) multiply 1-digit numbers by 10, e) appreciate arranging objects in groups of 3's, 4's, 5's, and 10's in real-life situations.	The learner is guided to: • use counters or other concrete objects to represent multiplication as repeated addition, • model multiplication as repeated addition using concrete objects, • practice using '×' sign in writing repeated addition sentences as multiplication, • work with peers and multiply 1-digit numbers by 1, 2, 3, 4, 5, • use locally available materials to model a multiplication chart and display it in the learning environment, • multiply 1-digit numbers by 10 to form multiples of 10, • play games involving multiplication using digital devices or other resources,	How is multiplication represented as repeated addition?

			● visit the local market to see how different fruits and other items are arranged in groups of 3's, 4's, 5's, or 10's for selling, and assist in grouping some of the items for sale.	

Core Competencies to be developed:

- Digital Literacy: learner uses digital devices to play games involving multiplication.
- Critical thinking and Problem-solving: learner uses locally available materials to model a multiplication chart.

Values:

- Responsibility: learner shares resources amicably as they model a multiplication chart.
- Patriotism: learner participates in community activities as they visit the local market and assist in grouping items for sale.

Pertinent and Contemporary Issues (PCIs):

- Financial Literacy: the learner visits the local market to see how fruits and other items are arranged in groups for selling.
- Community involvement: the learner visits the local market to assist in grouping items for sale.

Link to other learning areas:

The learner relates the skills used in improvising learning materials to waste management skills in Environmental Activities.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Numbers	**1.6 Division** (8 lessons) • *Division of numbers* • *Division sign*	By the end of the sub-strand, the learner should be able to: a) represent division as equal sharing up to the number 20 by 2, b) represent division as the equal grouping of numbers up to 25 by 3, c) use the '÷' sign in writing division statements. d) divide numbers up to 25 by 4 and 5 without a remainder, e) appreciate the application of division of numbers in real-life situations.	The learner is guided to: ● in teams, share a given number of objects equally by each picking one object at a time until all the objects are finished. Each learner to count how many objects he/she got, ● in teams, place several objects together, and let each team pick one item at a time until there is no object remaining, each team is to count the number of objects they picked, ● write division statements using the sign '÷', ● organise numbers up to 25 into groups of 4 or 5 without a remainder, ● play games involving division using digital devices or other resources with peers.	How can you share a given number of objects equally?

Core Competencies to be developed:

- Learning to Learn: learner learns to organise numbers up to 25 into groups of 4 or 5 without a remainder,
- Digital Literacy: learner uses digital devices to play games involving division.

Values: ● Love: the learner shares a given number of objects equally by each picking one object at a time until all the objects are finished. ● Unity: learner works harmoniously in teams as they place objects together.
Pertinent and Contemporary Issues (PCIs): ● Positive discipline: learner works harmoniously in teams as they place and share objects. ● Social Cohesion: learner plays games involving division using digital devices or other resources with peers.
Link to other learning areas: ● The learner relates the skills of writing division statements to functional writing in English Language Activities. ● The learner relates the concept of equal sharing to the concept of values in Religious Activities.

Strand	Sub-strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Numbers	**1.7 Fractions** (12 lessons) *Identifying fractions*	By the end of the sub-strand, the learner should be able to: a) identify a $\frac{1}{2}$ as part of a whole in different situations, b) identify a $\frac{1}{4}$ as part of a whole in different situations, c) apply fractions in day-to-day activities, d) appreciate use of fractions in daily life activities.	The learner is guided to: ● use papers, pencils, and a pair of scissors to make circular paper cut-outs while observing safety, ● fold the circular paper cut-outs into two equal parts and identify one of the parts as half of the whole written as $\frac{1}{2}$, ● make rectangular paper cut-outs and fold them into two equal parts to get half of a whole written as $\frac{1}{2}$, ● fold circular paper cut-outs to get 4 equal parts and identify one of the parts as a $\frac{1}{4}$ of a whole, ● practise making or sharing items in halves or quarters of a whole, ● play games involving fractions using digital devices or other resources.	How do we get a fraction from a whole?

Core Competencies to be developed: • Learning to Learn: the learner identifies halves and quarters as part of a whole in different situations. • Self-efficacy: learner practises making halves and quarters of a whole from paper cut-outs.
Values: • Responsibility: learner observes safety as they use scissors to make circular paper cut-outs. • Unity: learner collaborates with peers as they use digital devices to play games involving fractions.
Pertinent and Contemporary Issues (PCIs): • Life skills: learner uses fractions in day-to-day activities. • Self-esteem: learner practises making halves and quarters of a whole from paper cut-outs.
Link to other learning areas: The learner can relate the skills of making halves and quarters of a whole from paper cut-outs to pattern and modelling skills in Creative Activities.

SUGGESTED ASSESSMENT RUBRIC

Level / Indicator	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to identify numbers 1 to 100 in symbols.	The learner identifies numbers 1 to 100 in symbols and words accurately.	The learner identifies numbers 1 to 100 in symbols accurately.	The learner identifies numbers 1 to 70 in symbols accurately.	The learner identifies numbers 1 to 50 in symbols.
Ability to count numbers forward up to 100 and backward from number 50.	The learner counts numbers forward up to 100 and backward from number 50 accurately and fluently.	The learner counts numbers forward up to 100 and backward from number 50 accurately.	The learner counts numbers forward up to 70 and backward from number 30 accurately.	The learner counts numbers forward up to 50 and backward from number 20.
Ability to read and write numbers 1 to 100 in symbols and 1 to 20 in words.	The learner reads and writes numbers 1 to 100 in symbols and 1 to 20 in words correctly and proficiently.	The learner reads and writes numbers 1 to 100 in symbols and 1 to 20 in words correctly.	The learner reads and writes numbers 1 to 70 in symbols and 1 to 15 in words correctly.	The learner reads and writes numbers 1 to 70 in symbols and 1 to 10 in words.
Ability to identify the place value of numbers in ones and tens.	The learner identifies the place value of numbers in ones, tens, and hundreds correctly.	The learner identifies the place value of numbers in ones and tens correctly.	The learner identifies the place value of numbers in ones or tens correctly.	The learner identifies the place value of numbers in ones or tens partially correctly.

Ability to work out missing numbers in number patterns up to 100.	The learner works out missing numbers in number patterns up to 100 correctly and systematically.	The learner works out missing numbers in number patterns up to 100 correctly.	The learner works out missing numbers in number patterns up to 100 partially correctly.	The learner works out missing numbers in number patterns up to any number less than 100 partially correctly.
Ability to add a 2-digit number to a 2-digit number without and with regrouping, with a sum not exceeding 100.	The learner adds a 2-digit number to a 2-digit number without and with regrouping, with a sum not exceeding 100 correctly and systematically.	The learner adds a 2-digit number to a 2-digit number without and with regrouping, with a sum not exceeding 100 correctly.	The learner adds a 2-digit number to a 2-digit number without or with regrouping, with a sum not exceeding 100 correctly.	The learner adds a 2-digit number to a 1-digit number without or with regrouping, with a sum not exceeding 100.
Ability to work out missing numbers in patterns involving addition and subtraction of whole numbers up to 100.	The learner works out missing numbers in patterns involving addition and subtraction of whole numbers up to 100 correctly and systematically.	The learner works out missing numbers in patterns involving addition and subtraction of whole numbers up to 100 correctly.	The learner works out missing numbers in patterns involving the addition and subtraction of whole numbers up to 100 partially correctly.	The learner works out missing numbers in patterns involving the addition or subtraction of whole numbers up to any number less than 100 partially correctly.

Ability to subtract a 2-digit number from a 2-digit number without and with regrouping.	The learner subtracts a 2-digit number from a 2-digit number without and with regrouping correctly and systematically.	The learner subtracts a 2-digit number from a 2-digit number without and with regrouping correctly.	The learner subtracts a 2-digit number from a 2-digit number without and with regrouping partially correctly.	The learner subtracts a 2-digit number from a 2-digit number without or with regrouping partially correctly.
Ability to multiply 1-digit numbers by 1, 2, 3, 4, 5 and 10.	The learner multiplies 1-digit numbers by 1, 2, 3, 4, 5, and 10 accurately and systematically.	The learner multiplies 1-digit numbers by 1, 2, 3, 4, 5 and 10 accurately.	The learner multiplies 1-digit numbers by any 3 to 5 of; 1, 2, 3, 4, 5, or 10 accurately.	The learner multiplies 1-digit numbers by any 2 of; 1, 2, 3, 4, 5, or 10.
Ability to represent division as equal sharing and grouping up to number 20 by 2.	The learner represents division as equal sharing and grouping up to the number 20 by 2 accurately and systematically.	The learner represents division as equal sharing and grouping up to the number 20 by 2 accurately.	The learner represents division as equal sharing or grouping up to the number 15 by 2 accurately.	The learner represents division as equal sharing or grouping up to number 10 by 2.
Ability to divide numbers up to 25 by 4 and 5 without a remainder.	The learner divides numbers up to 25 by 4 and 5 without a remainder accurately and systematically.	The learner divides numbers up to 25 by 4 and 5 without a remainder accurately.	The learner divides numbers up to 20 by 4 or 5 without a remainder accurately.	The learner divides numbers up to 15 by 4 or 5 without a remainder.
Ability to identify a $\frac{1}{2}$ and a $\frac{1}{4}$ as part of a whole.	The learner identifies a $\frac{1}{2}$ and a $\frac{1}{4}$ as part of a whole correctly and systematically.	The learner identifies a $\frac{1}{2}$ and a $\frac{1}{4}$ as part of a whole correctly.	The learner identifies a $\frac{1}{2}$ and a $\frac{1}{4}$ as part of a whole partially correctly.	The learner identifies a $\frac{1}{2}$ or an as part of a with partially correctly.

STRAND 2.0 MEASUREMENT

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Measurement	**2.1 Length** (6 lessons) *Measuring length*	By the end of the sub-strand, the learner should be able to: a) measure length using fixed units, b) identify the metre as a unit of measuring length, c) measure length in metres, d) appreciate measuring length using fixed units in real-life situations.	The learner is guided to: • use sticks of equal length to measure the lengths of different objects and record the measurements, • use sticks of different lengths to measure length, including 1-metre sticks, • in teams, use locally available materials to make 1-metre sticks and use them to measure the length of various objects within the classroom and record the measurements, • measure the length of different objects at home, record the measurements, and discuss them with peers in school.	How is the length of an object measured?

Core Competencies to be developed:

- Critical thinking and Problem-solving: learner makes 1-metre sticks and uses them to measure the length of various objects.
- Self-efficacy: the learner measures the length of different objects at home, records the measurements, and discusses them with peers in school.

Values:

- Responsibility: the learner uses locally available materials to make 1-metre sticks and use them to measure the length of various objects.
- Unity: the learner works in teams to make 1-metre sticks and use them to measure the length of various objects.

Pertinent and Contemporary Issues (PCIs):

- Self-esteem: the learner measures the length of different objects at home, records the measurements, and discusses them with peers in school.
- Creative thinking: learner uses locally available materials to make a 1-metre stick.

Link to Other Learning Areas:

The learner can relate the concept of using objects within the classroom and at home for learning to the concept of resources in the environment in Environmental Activities.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Measurement	**2.2 Mass** (6 lessons) *Measuring mass*	By the end of the sub-strand, the learner should be able to: a) measure mass using fixed units, b) identify the kilogram as a unit of measuring mass, c) measure the mass of different objects in kilograms, d) appreciate measuring mass using fixed units in real-life situations.	The learner is guided to: ● collect safe materials from the environment for measuring mass, ● work with peers to use locally available materials to make an improvised a beam balance, ● in teams, use items of the same mass and an improvised beam balance to measure different masses, record and discuss the results with peers, ● use an item equivalent to a 1-kilogram mass and a beam balance to measure the mass of different objects in kilograms accurately, ● visit a shop or market and assist vendors in measuring the mass of items such as beans, maize, rice, and flour using fixed units, for example, an empty 1-kilogram container, ● measure the mass of different items in kilograms using a 1-kilogram container, ● play games using digital devices or other resources involving mass in kilograms.	Why is it important to know the mass of an object?

Core Competencies to be developed:

- Communication and Collaboration: learner in teams, uses items of the same mass and an improvised beam balance to measure different masses and discuss results.
- Creativity and imagination: learner works with peers to use locally available materials to make an improvised a beam balance.

Values:

- Patriotism: the learner assists vendors in measuring the masses of items such as beans, maize, rice, and flour using fixed units.
- Responsibility: learner improvises beam balance to measure different masses.

Pertinent and Contemporary Issues (PCIs):

- Creative thinking: learner improvises beam balance to measure different masses.
- Community involvement: learner visits a shop or market and assists vendors to measure the masses of items using fixed units.

Link to Other Learning Areas:

- The learner can relate the concept of visiting a shop or market to the concept of our market in Environmental Activities.
- The learner can relate the skills of using locally available materials to improvise a beam balance to the skills of identifying resources in the environment in Environmental Activities.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Measurement	**2.3 Capacity** (8 lessons) *Measuring capacity*	By the end of the sub-strand, the learner should be able to: a) measure capacity using fixed units, b) identify the litre as a unit of measuring capacity, c) measure capacity in litres, d) appreciate measuring capacity in litres using improvised containers in real-life situations.	The learner is guided to: ● collect containers of different sizes for use in measuring capacity, ● use small containers of equal capacity to fill bigger containers of the same capacity but different shapes with water, count the number of small containers used to fill the big containers, ● use 1-litre containers to fill bigger containers with water, and count the number of litres used to fill the big containers. Use water properly to avoid wetting floors. ● in teams, discuss and measure the capacity of different containers in litres accurately, ● participate in activities involving measuring liquids such as milk and water using 1-litre bottles.	1. Why do containers have different capacities? 2. Which commodities can be measured in terms of litres?

Core Competencies to be developed:

- Self-efficacy: learner uses 1-litre containers to fill bigger containers with water and counts the number of litres used to fill the bigger containers.
- Communication and Collaboration: learner in teams, discusses and measures the capacity of different containers in litres.

Values:

- Unity: learner participates in activities involving measuring liquids such as milk and water using 1-litre bottles.
- Responsibility: The learner collects containers of different sizes for use in measuring capacity.

Pertinent and Contemporary Issues (PCIs):

- Safety issues: the learner uses water carefully to avoid wet floors as they use 1-litre containers to fill big containers with water.
- Self-esteem: learner discusses and measures the capacity of different containers in litres confidently.

Link to other Learning Areas:

The learner can relate the skills used in discussion to speaking and listening skills in English Language Activities.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Measurement	**2.4 Time** (10 lessons) • *Months of the year* • *Days in a month* • *Calendar* • *Minute and hour hand* • *Reading and writing time*	By the end of the sub-strand, the learner should be able to: a) relate the months of the year to special occasions, b) recite the number of days in each month of the year, c) read the calendar in terms of day and date, d) measure time using arbitrary units, e) identify the minute and hour hand in the clock face, f) read time by the hour from the clock face, g) write time by the hour shown by an analogue and digital clock,	The learner is guided to: • discuss special occasions that take place in different months of the year, • sing songs and rhymes related to the number of days in the months of the year, • discuss how to read, tell, and write dates from the calendar. • discuss and relate time by hour using the length of a shadow of an object such as a tree in the environment, • discuss places where clocks are displayed and how they look, • observe a clock face and discuss the minute and hour hand, • discuss how to read, tell and write time by the hour using both the analogue and digital clock,	How do we use calendars and clocks to track time?

		h) appreciate keeping time in day-to-day activities.	• discuss the importance of keeping time for different activities.	

Core Competencies to be developed:

- Citizenship: the learner discusses special occasions that take place in different months of the year.
- Self-efficacy: the learner reads the calendar in terms of day and date, and measures time using arbitrary units.

Values:

- Patriotism: the learner discusses special occasions that take place in different months of the year.
- Integrity: the learner discusses the importance of keeping time in different activities.

Pertinent and Contemporary Issues (PCIs):

- Global Citizenship: the learner discusses special occasions that take place in different months of the year.
- Life Skills: learner discusses and relates time by the hour using the length of a shadow of an object such as a tree in the environment.

Link to other learning areas:

The learner can relate the skills of telling time to the concept of weather and sky, and day and night in Environmental Activities.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested learning experiences	Suggested Key Inquiry Question(s)
2.0 Measurement	**2.5 Money** (10 lessons) • *Kenyan currency* • *Counting money* • *Addition in money*	By the end of the sub-strand, the learner should be able to: a) identify Kenyan currency coins and notes up to sh.100, b) count the number of sh. 10 and sh. 20 coins in different situations, c) count the number of sh. 50 and sh. 100 notes in different situations, d) add two denominations of money with a sum not exceeding sh. 100, e) use money to buy up to 3 items without balance, f) appreciate the use of money in buying items.	The learner is guided to: • recognise and sort out Kenyan currency coins and notes up to sh.100, • in teams, make sh. 10 and sh. 20 coins paper cut-outs, sort and count the number of sh. 10 and sh. 20 coins paper cut-outs, • work collaboratively, to make sh. 50 and sh. 100 notes paper cut-outs, sort and count the number of sh. 50 and sh. 100 notes paper cut-outs, • add two denominations of money with a sum not exceeding sh. 100, • with peers, role play buying up to 3 items without balance from the model shop in the classroom, • record a video during a role play of classroom shopping activities.	How different are Kenyan currency denominations?

Core Competencies to be developed:

- Collaboration: learner in teams, makes sh. 10 and sh. 20 coins paper cut-outs, sorts and counts the number of sh. 10 and sh. 20 coins paper cut-outs.
- Digital Literacy: the learner records a video during a role play of classroom shopping activities.

Values:

- Patriotism: learner recognises and sorts out Kenyan currency coins and notes up to sh.100.
- Responsibility: learner takes up assigned roles as they role-play classroom shopping activities.

Pertinent and Contemporary Issues (PCIs):

- Financial Literacy: learner role plays buying up to 3 items without balance from the model shop in the classroom.
- Citizenship: the learner recognises and sorts out Kenyan currency coins and notes.

Link to other learning areas:

The learner can relate the skills of using money to buy items to the concept of our market in Environmental Activities.

SUGGESTED ASSESSMENT RUBRIC

LEVEL / INDICATOR	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to measure the length in metres, mass in kilograms, and capacity in litres.	The learner measures length in metres, mass in kilograms, and capacity in litres accurately and systematically.	The learner measures length in metres, mass in kilograms, and capacity in litres accurately.	The learner measures any 2 of; length in metres, mass in kilograms, and capacity in litres accurately.	The learner measures any 1 of; length in metres, mass in kilograms, or capacity in litres.
Ability to read the calendar in terms of day and date.	The learner reads the calendar in terms of day, month, and year accurately and fluently.	The learner reads the calendar in terms of day, month, and year accurately.	The learner reads the calendar in terms of any 2 of; day, month, and year accurately.	The learner reads the calendar in terms of any; day, month, or year.
Ability to read and write time by the hour from the clock face.	The learner reads and writes time by the hour from the clock face accurately and fluently.	The learner reads and writes time by the hour from the clock face accurately.	The learner reads or writes time by the hour from the clock face accurately.	The learner reads or writes time by the hour from the clock face partially accurately.
Ability to identify Kenyan currency coins and notes up to sh.100.	The learner identifies Kenyan currency coins to sh. 40 and notes up to sh.100 accurately and names their features.	The learner identifies Kenyan currency coins to sh. 40 and notes up to sh.100 accurately	The learner identifies Kenyan currency coins up to sh. 40 or notes up to sh.100 accurately.	The learner identifies Kenyan currency coins up to sh. 20 or sh.50 note.

Ability to count the number of sh.10 and sh.20 coins and sh.50 and sh.100 notes.	The learner counts the number of sh.10 and sh.20 coins and sh.50 and sh.100 notes correctly and fluently.	The learner counts the number of sh.10 and sh.20 coins and sh.50 and sh.100 notes correctly.	The learner counts the number of any 3 of; sh.10 and sh.20 coins, sh.50 and sh.100 notes correctly.	The learner counts the number of any 2 or 1 of; sh.10 or sh.20 coins, sh.50 or sh.100 notes.
Ability to add 2 denominations of money with a sum not exceeding sh. 100.	The learner adds 2 denominations of money with a sum not exceeding sh. 100 correctly and consistently.	The learner adds 2 denominations of money with a sum not exceeding sh. 100 correctly.	The learner adds 2 denominations of money with a sum not exceeding Sh. 70 correctly.	The learner adds 2 denominations of money with a sum not exceeding sh. 50.

STRAND 3.0 GEOMETRY

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
3.0 Geometry	**3.1 Lines** (5 lessons) • *Straight lines* • *Curved lines*	By the end of the sub-strand, the learner should be able to: a) model straight lines in different ways, b) draw straight lines in different ways, c) model curved lines in different ways, d) draw curved lines in different situations, e) recognise the use lines in real-life situations.	The learner is guided to: • safely model straight lines using sticks, plasticine/clay, or papier mache and strings, • safely model curved lines using plasticine/clay, or papier mache and strings, • model straight and curved lines by holding their hands in different ways; upward, horizontal, diagonal, • draw straight and curved lines on manila papers and books using pencils, crayons rulers, or sticks, • use digital devices or other resources to draw lines and share with peers.	How are lines used in real life?
Core Competencies to be developed: • Self-efficacy: learner draws straight and curved lines on manila papers and books. • Digital Literacy: learner uses digital devices to draw lines.				

Values: ● Responsibility: the learner takes care of materials used for drawing straight and curved lines. ● Love: learner portrays a caring attitude as they use digital devices to draw lines and share with peers.
Pertinent and Contemporary Issues (PCIs): ● Creative thinking: learner safely models curved lines using plasticine/clay or papier mache and strings. ● Friendship formation: the learner uses digital devices or other resources to draw lines and share with peers.
Links to other learning areas: The learner can relate the skills used in drawing straight and curved lines to drawing skills in Creative Activities.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
3.0 Geometry	**3.2 Shapes** (5 lessons) *Drawing shapes and patterns*	By the end of the sub-strand, the learner should be able to: a) identify different shapes in the environment, b) draw shapes in different ways, c) draw patterns involving different shapes, d) appreciate the use of shapes in forming patterns in fabrics.	The learner is guided to: ● recognise and name different shapes in the environment (triangles, rectangles, squares, circles, and ovals), ● discuss types of lines in different shapes (triangles, rectangles, squares, circles, and ovals) ● in turn, name different shapes of objects or on surfaces in their classroom, ● draw triangles, rectangles, squares, circles, and ovals on manila papers and display them in the learning environment, ● make patterns using triangles, rectangles, squares, circles, and ovals, colour them and share them with peers, ● play games involving pattern-making using digital devices or other resources with peers.	How can patterns be made using shapes?
Core Competencies to be developed: • Creativity and Imagination: learner makes patterns using triangles, rectangles, squares, circles, and ovals, and colour them. • Learning to Learn: learner recognises and names different shapes in the environment				

Values:

- Unity: the learner discusses types of lines in different shapes and in turn, names different shapes of objects in their classroom.
- Social Justice: learner gives others equal opportunity as they in turn name different shapes of objects in their classroom.

Pertinent and Contemporary Issues (PCIs):

- Self-esteem: the learner draws triangles, rectangles, squares, circles, and ovals on manila papers and displays them in the learning environment.
- Social Cohesion: learner plays games involving pattern-making using digital devices or other resources with peers.

Link to other learning areas:

The learner can relate the skills used in drawing shapes to drawing skills in Creative Activities.

SUGGESTED ASSESSMENT RUBRIC

LEVEL / INDICATOR	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to model and draw straight and curved lines.	The learner models and draws straight lines correctly and creatively.	The learner models and draws straight lines correctly.	The learner models or draws straight lines correctly.	The learner models or draws straight lines partially correctly.
Ability to model and draw curved lines.	The learner models and draws curved lines correctly and creatively.	The learner models and draws curved lines correctly.	The learner models or draws curved lines correctly.	The learner models or draws curved lines partially correctly.
Ability to identify and draw shapes (triangles, rectangles, squares, circles, and ovals).	The learner identifies and draws 5 shapes (triangles, rectangles, squares, circles, and ovals) accurately and consistently.	The learner identifies and draws 5 shapes (triangles, rectangles, squares, circles, and ovals) accurately.	The learner identifies and draws any 4 of the 5 shapes (triangles, rectangles, squares, circles, and ovals) accurately.	The learner identifies or draws less than 4 shapes (triangles, rectangles, squares, circles, and ovals).
Ability to draw patterns involving shapes (triangles, rectangles, squares, circles, and ovals).	The learner draws patterns involving 5 shapes (triangles, squares, circles, and ovals) accurately and creatively.	The learner draws patterns involving 5 shapes (triangles, rectangles, squares, circles, and ovals) accurately.	The learner draws patterns involving any 4 of the 5 shapes (triangles, rectangles, squares, circles, and ovals) accurately.	The learner draws patterns involving less than 4 shapes (triangles, rectangles, squares, circles, and ovals).

SUGGESTED LEARNING RESOURCES

STRANDS	SUB -STRANDS	RESOURCES
NUMBERS	NUMBER CONCEPT	Counters such as bottle tops, marbles, sticks, stones, grains
	WHOLE NUMBERS	Bottle tops, marbles, sticks, stones, grains, a number line drawn on the ground/floor
	ADDITION	Bottle tops, marbles, stones, sticks, grains, place value chart, abacus, basic addition facts table, a number line drawn on the ground/floor
	SUBTRACTION	Bottle tops, marbles, sticks, stones, grains, basic addition facts table, a number line drawn on the ground/floor
	MULTIPLICATION	Bottle tops, marbles, stones, grains, number lines drawn on the ground/floor, multiplication table
	DIVISION	Bottle tops, marbles, sticks, stones, grains, multiplication tables
	FRACTIONS	Circular and rectangular cut-outs, pair of scissors
MEASUREMENT	LENGTH	Pencils, sticks, rulers, strings, ropes
	MASS	Items of different masses such as books, stones, pieces of wood, items of the same mass, beam balance
	CAPACITY	Containers of different sizes, 1-litre containers, water, soil, sand
	TIME	Charts with the number of days in each month and months of the year in order, clock face both analogue and digital
	MONEY	Money in coins and notes sh.1, sh.5, sh.10, sh.20, sh.40, sh.50, sh.100, Model classroom shop
GEOMETRY	LINES	Sticks, clay, plasticine, strings, ropes
	SHAPES	Cut-outs of rectangles, circles, triangles, ovals, and squares of different sizes

NOTE

The following **ICT** devices may be used in the teaching/learning of Mathematics at this level:

- Learner digital devices (LDD),
- Teacher digital devices (TDD),
- Mobile phones,
- Digital clocks,
- Television sets,
- Videos,
- Cameras,
- Projectors,
- Radios,
- DVD players
- CD's,
- Scanners,
- Internet among others.

Suggested Assessment Methods and Tools

1. Written tests and quizzes
2. Rating scales
3. Projects
4. Observation Schedules
5. Portfolios
6. Assessment Rubric
7. Questionnaire